

Johnson
Controls



WILMAR

WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726



CUSTOMER P.O NO : 4502248834

D	AS BUILT	1	DRAWN BY:	CWY
		07.08.2018		
C	SHOP DRAWING	2	CHECKED BY:	ELVIN
		24.07.2018		
B	SEND FOR APPROVAL	3	VERIFIED BY:	JAC KER
		12.06.2018		
LEV.	REVISION	REV.	SCALE	N.T.S
		DATE		

USER :	Wilmar
PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	COVER SHEET

MANUFACTURER : SWIFT ^{energy} SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
PROJECT NO : A1331	JOB NO : -
SHT.NO. : 1	

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E	A1331/ 101		SINGLE LINE DIAGRAM													N.T.S		1/8/2018		CWY																				
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	A1331/ 201		POWER & CONTROL CIRCUIT - CONDENSER PUMP													N.T.S		2/8/2018		CWY																				
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	CUSTOMER P.O NO :				12.06.2018					PROJECT : WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726									PROJECT NO : A1331			JOB NO : -																		
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					DATE	SCALE																																		

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	D	AS BUILT	1 07.08.2018	DRAWN BY: CWY	USER : Wilmar	MANUFACTURER :  SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
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	B	SEND FOR APPROVAL	3 12.06.2018	VERIFIED BY: JAC KER			
	CUSTOMER P/O NO : 4502248834	LEV.	REVISION	REV. DATE	SCALE N.T.S		PROJECT : WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
					TITLE : TABLE OF CONTENTS		SHT.NO. : 3

Guidelines for Cabling and Routing of Signal, Communication and Power Cables

Signal Cables

1. It is recommended to use screened signal cables, eg. 4-20mA circuits, 24 VDC circuits and other analogue circuits.
2. Terminate the screen to earth in only one end of signal and controls cables avoiding pigtails
3. It is recommended to terminate the screen to earth in the end by the electrical panel/compressor controller.

Power Cables

1. When using a VSD controlled motor starter, always use screened motor cable.
2. Connect the screen to earth in both ends of motor cables (both in motor starter panel and in motor terminal box).

Routing of Cables (Refer Figure 1)

1. It is important to separate the signal (e.g. 4-20 mA signals) and communication (e.g. Profibus communication) cables from the motor and mains cables, especially when using VSD.
2. The distance between the communication/signal cables and motor and mains cables must be at least 200mm. The distance between motor and mains cables must also be at least 200 mm. If it is necessary that cables cross each other they should always cross in a 90° angle.

IDENTIFICATION OF WIRING INSIDE CONTROL CABINET

RED	PHASE 1 CONDUCTORS
YELLOW	PHASE 2 CONDUCTORS
BLUE	PHASE 3 CONDUCTORS
BLACK	NEUTRAL CONDUCTORS
GREEN-AND-YELLOW	POWER EARTH CONDUCTORS (PE)
RED	230VAC CONTROL CIRCUITS (LIVE)
BLACK	230VAC CONTROL CIRCUITS (NEUTRAL)
BLUE	24VDC CONTROL CIRCUITS (POSITIVE)
WHITE	24VDC CONTROL CIRCUITS (NEGATIVE)
DEPEND ON AC OR DC CARRY	DRY CONTACT CABLE
GREEN	INSTRUMENT EARTH CONDUCTORS (IE)

IDENTIFICATION OF COMPONENTS, PAGES AND CONTACT

FOR COMPONENT :	K5	FOR TERMINAL GROUP, CABLE MARKER, CONTACT NO. :	
	COMPONENT NUMBER		TERMINAL GROUPING
	COMPONENT TYPE		
FOR PAGE :	/204.17		TERMINAL NUMBER
	COLUMN 17		CABLE MARKER
	PAGE 204		CONTACT NUMBER
FOR CONTACT :			FOR RELAY
	13 / 204.19		
	21 / 204.17		
	NORMALLY OPEN		
	NORMALLY CLOSED		

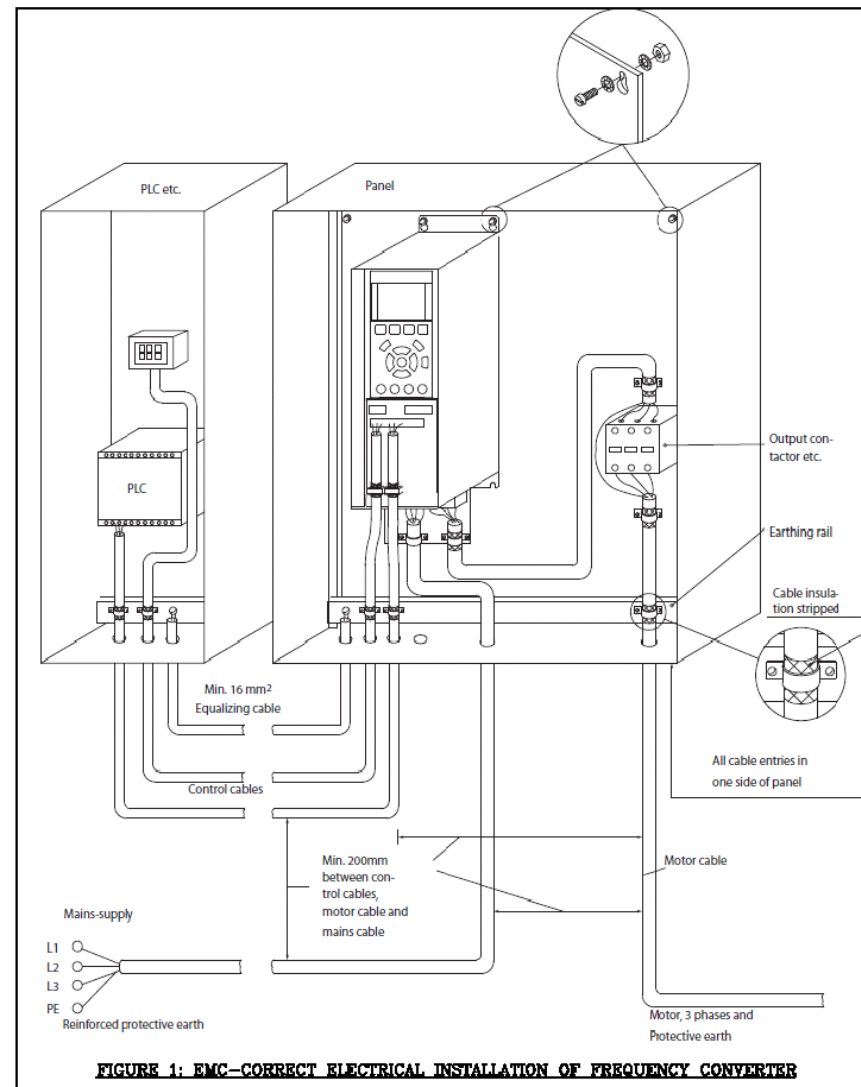


FIGURE 1: EMC-CORRECT ELECTRICAL INSTALLATION OF FREQUENCY CONVERTER

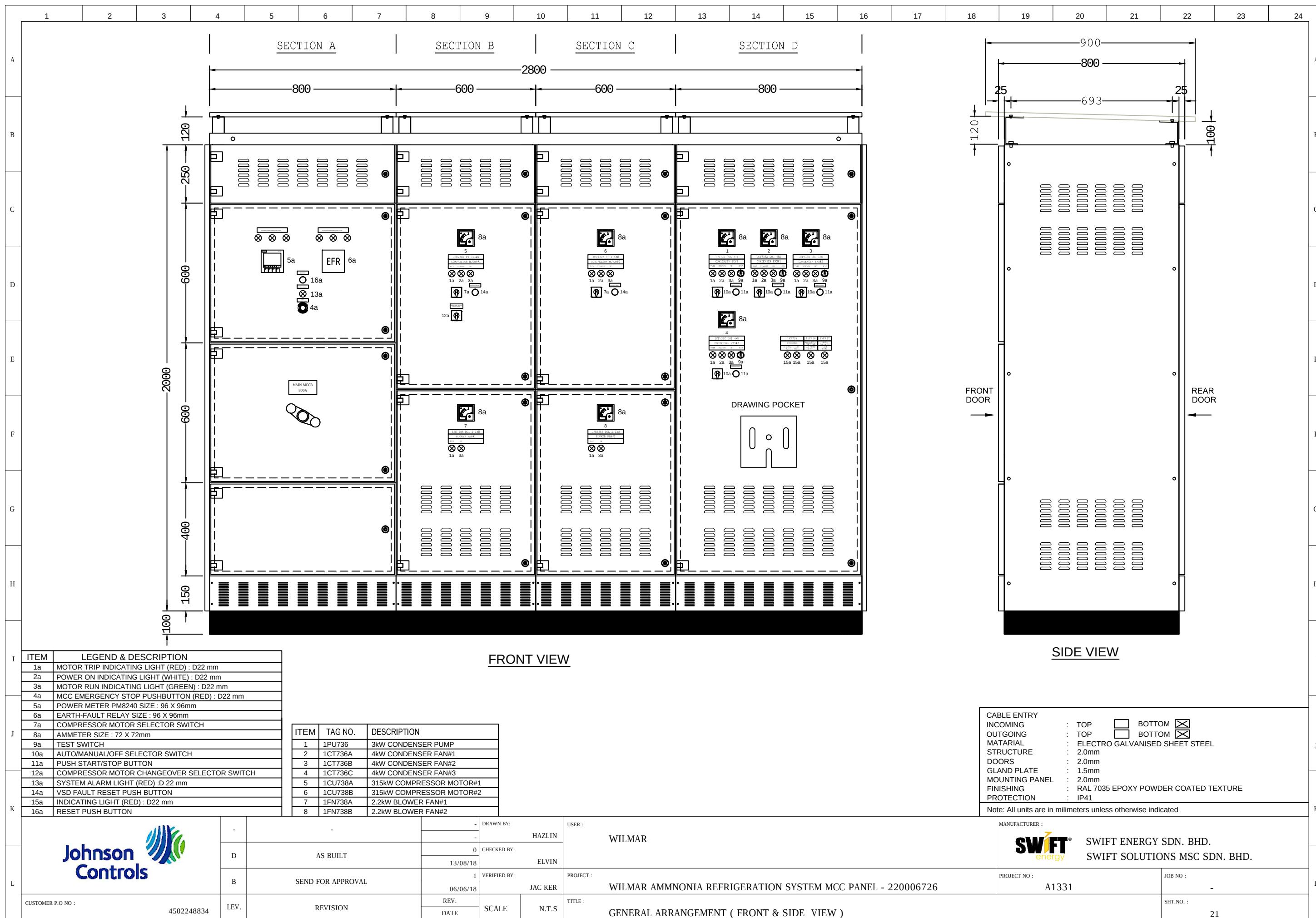


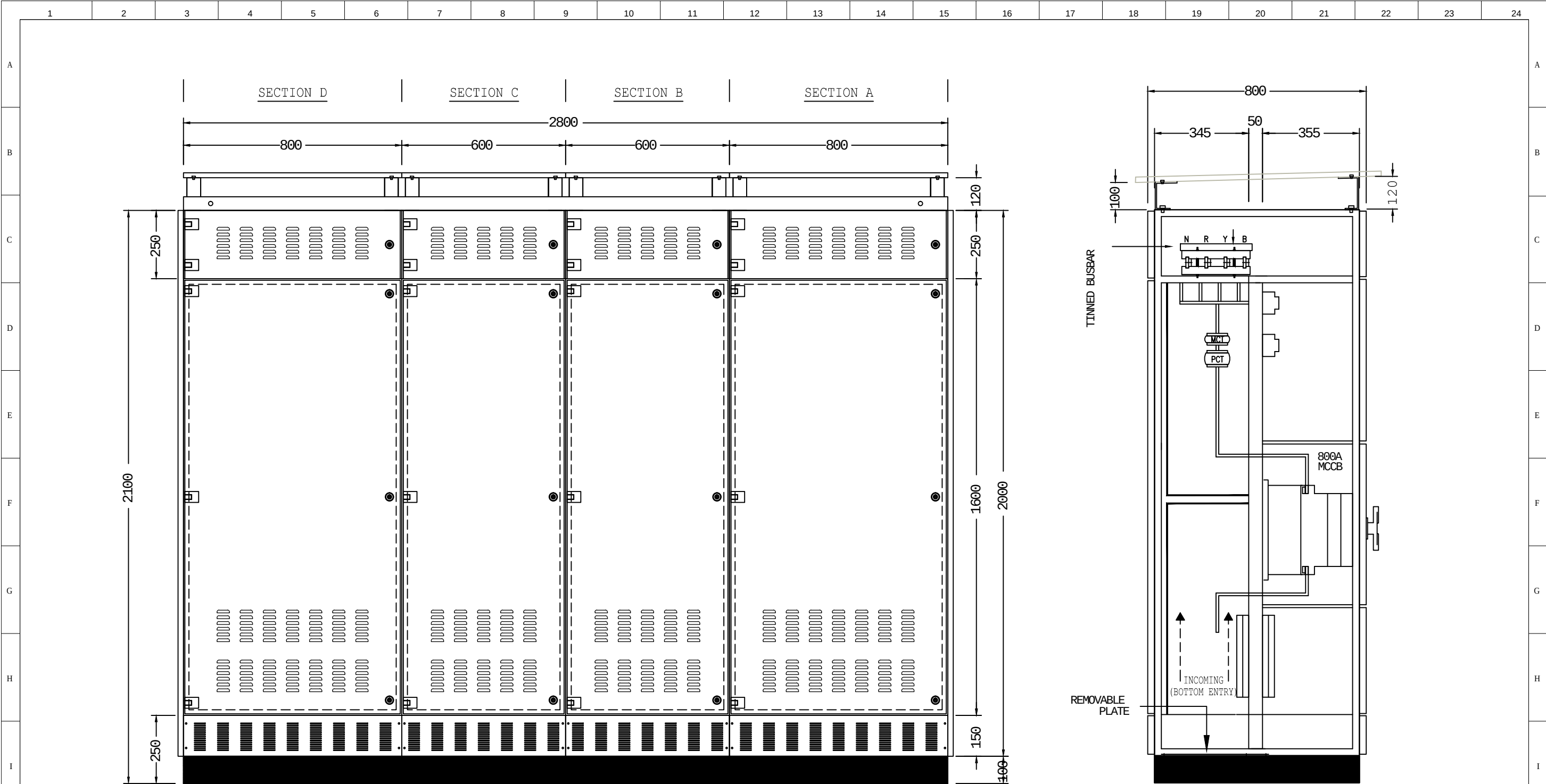
CUSTOMER P.O. NO : 4502248834

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LEV.	REVISION	REV.	SCALE	N.T.S
		DATE		

USER :	WILMAR
PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	TECHNICAL NOTES

MANUFACTURER :	SWIFT ENERGY SDN. BHD.
PROJECT NO :	A1331
JOB NO :	-
SHT.NO. :	20

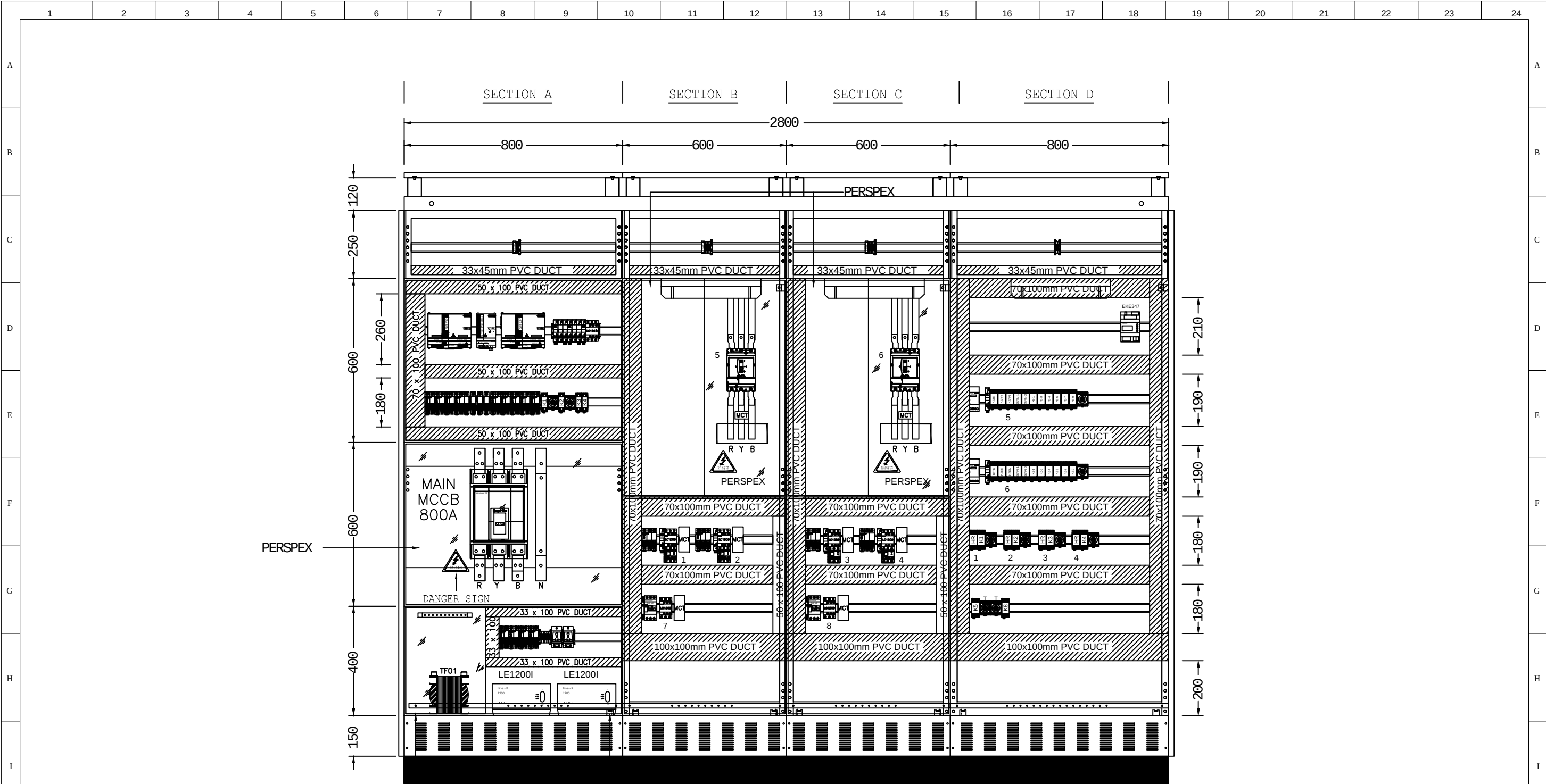




REAR VIEW

SECTIONAL SIDE VIEW

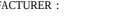
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					HAZLIN				 SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
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				13/08/18	ELVIN					
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			06/06/18	JAC KER						
CUSTOMER P.O NO : 4502248834		LEV.	REVISION	REV.	SCALE	N.T.S	TITLE :		SHT.NO. : 26	
				DATE			GENERAL ARRANGEMENT (REAR & SECTIONAL SIDE VIEW)			



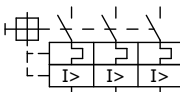
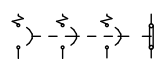
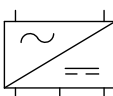
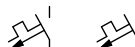
INTERNAL FRONT VIEW

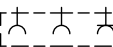
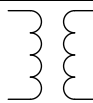
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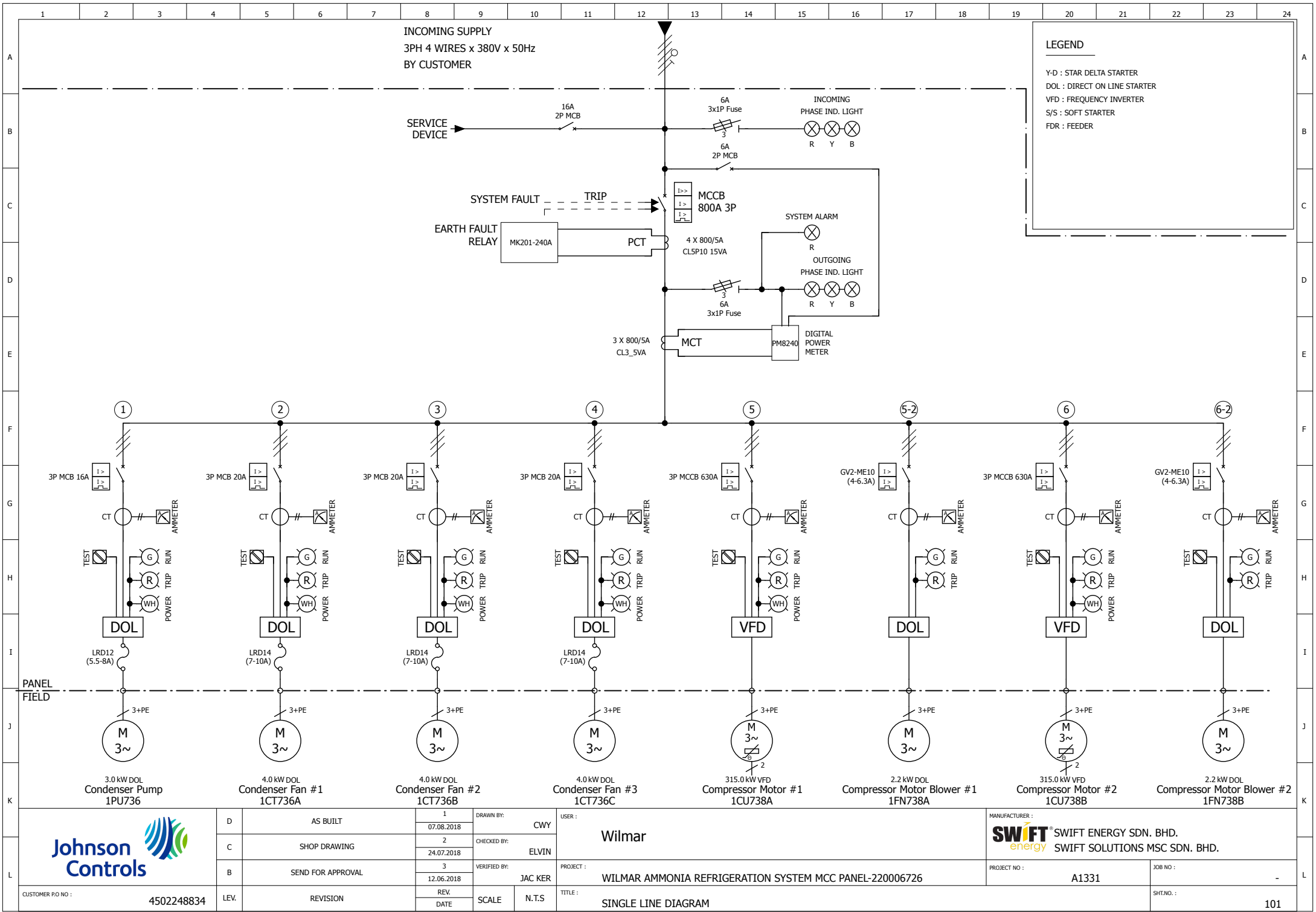
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CUSTOMER P.O NO :	LEV.	REVISION	REV.	SCALE	N.T.S	TITLE :			SHT.NO. :	
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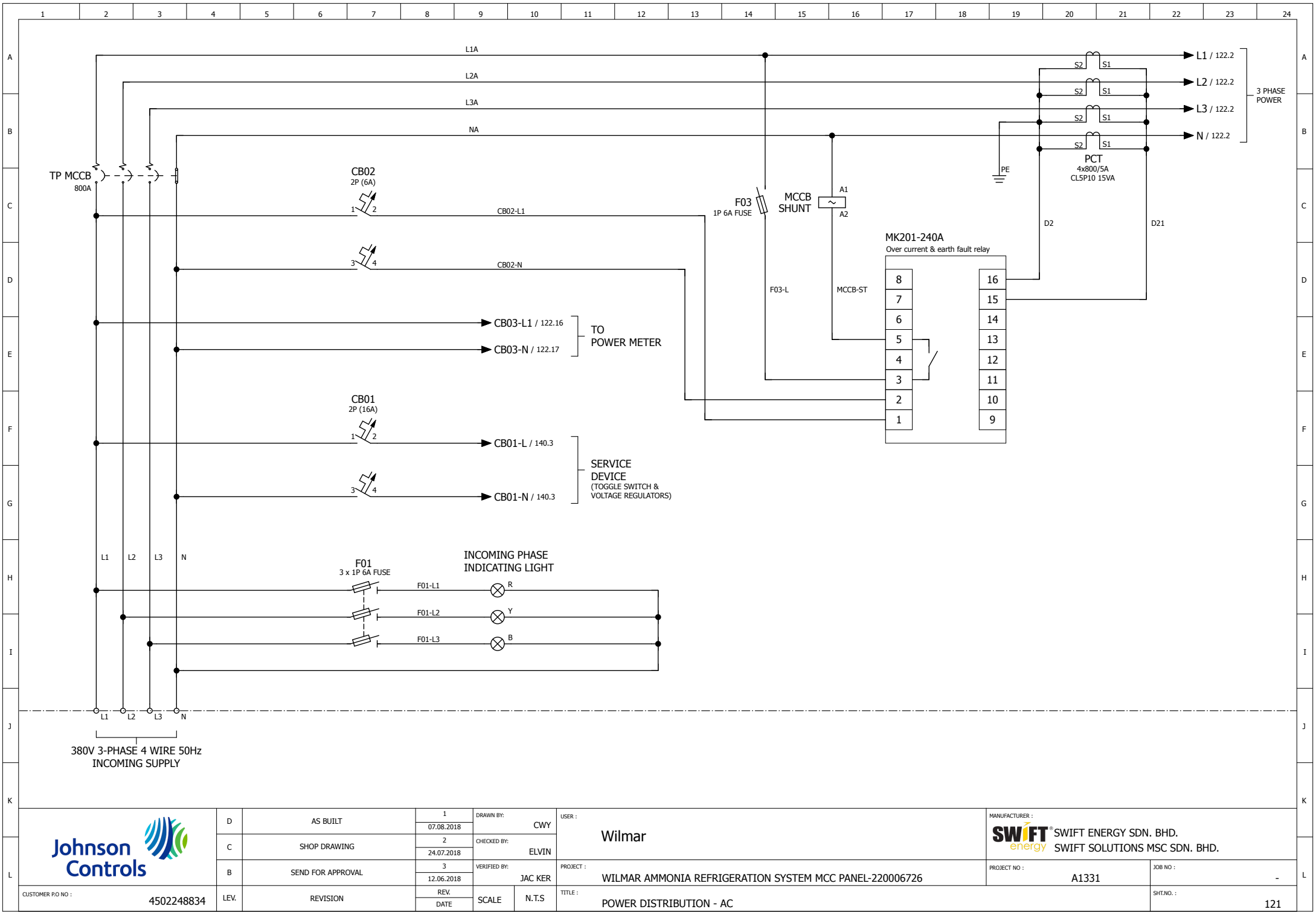
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SYMBOLS	DESCRIPTION
	MAGNETIC MANUAL STARTER
	CONTACTOR
	NORMALLY OPENED CONTACT
	NORMALLY CLOSED CONTACT
	RELAY / CONTACTOR COIL
	FUSE
	MCCB
	PILOT LIGHT / LIGHTING
	POWER SUPPLY
	CIRCUIT BREAKER
	TERMINAL

SYMBOLS	DESCRIPTION
	CIRCUIT BREAKER
	LIGHTING
	VOLTAGE REGULATOR
	NORMALLY OPEN LIMIT SWITCH
	OFF DELAY TIMER
	24VDC RELAY
	EMERGENCY STOP PUSH BUTTON
	TRANSFORMER
	EARTH

	D	AS BUILT	1 07.08.2018	DRAWN BY: CWY	USER : Wilmar	MANUFACTURER : SWIFT [®] SWIFT ENERGY SDN. BHD. energy SWIFT SOLUTIONS MSC SDN. BHD.	
	C	SHOP DRAWING	2 24.07.2018	CHECKED BY: ELVIN		PROJECT NO : A1331	
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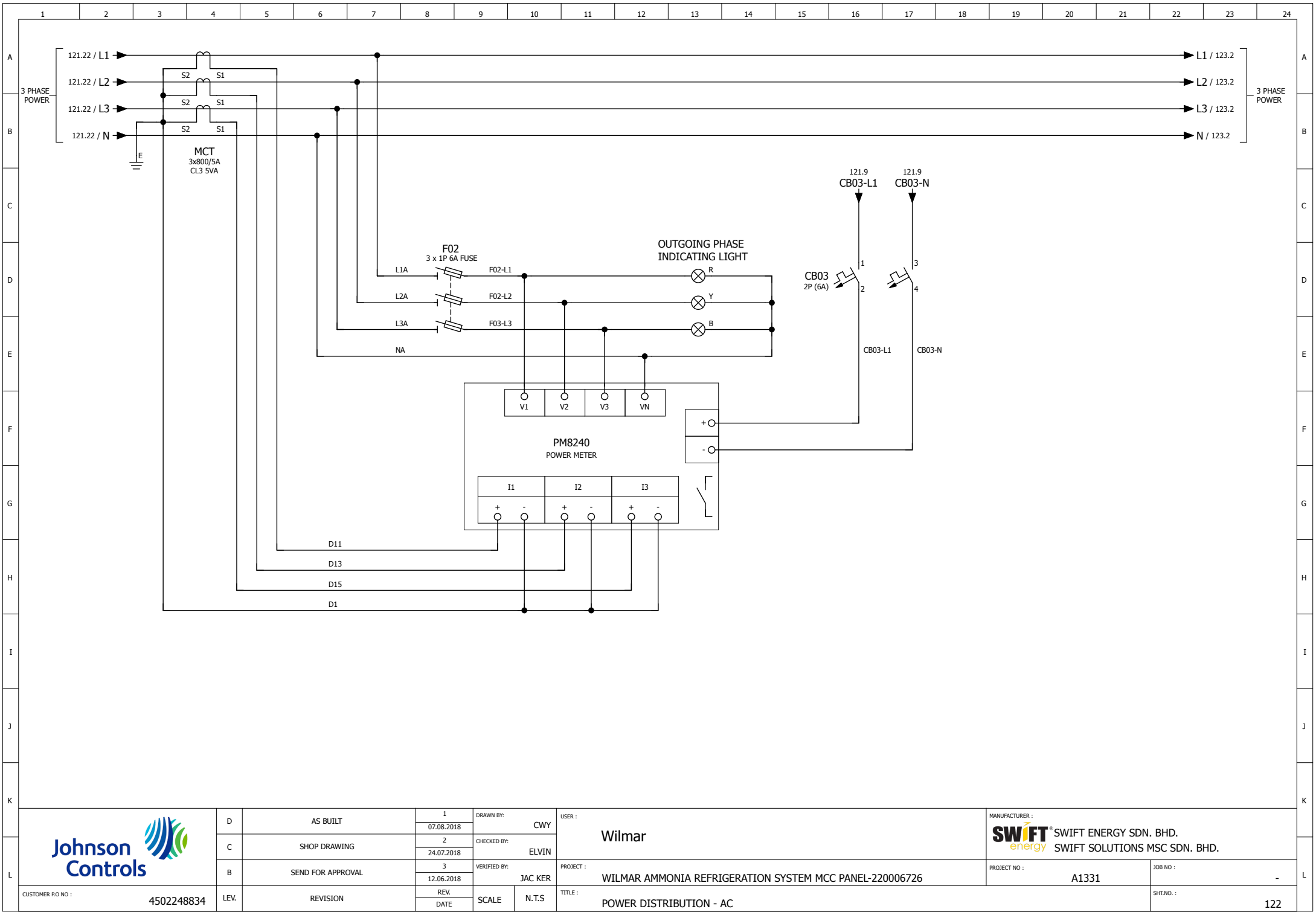




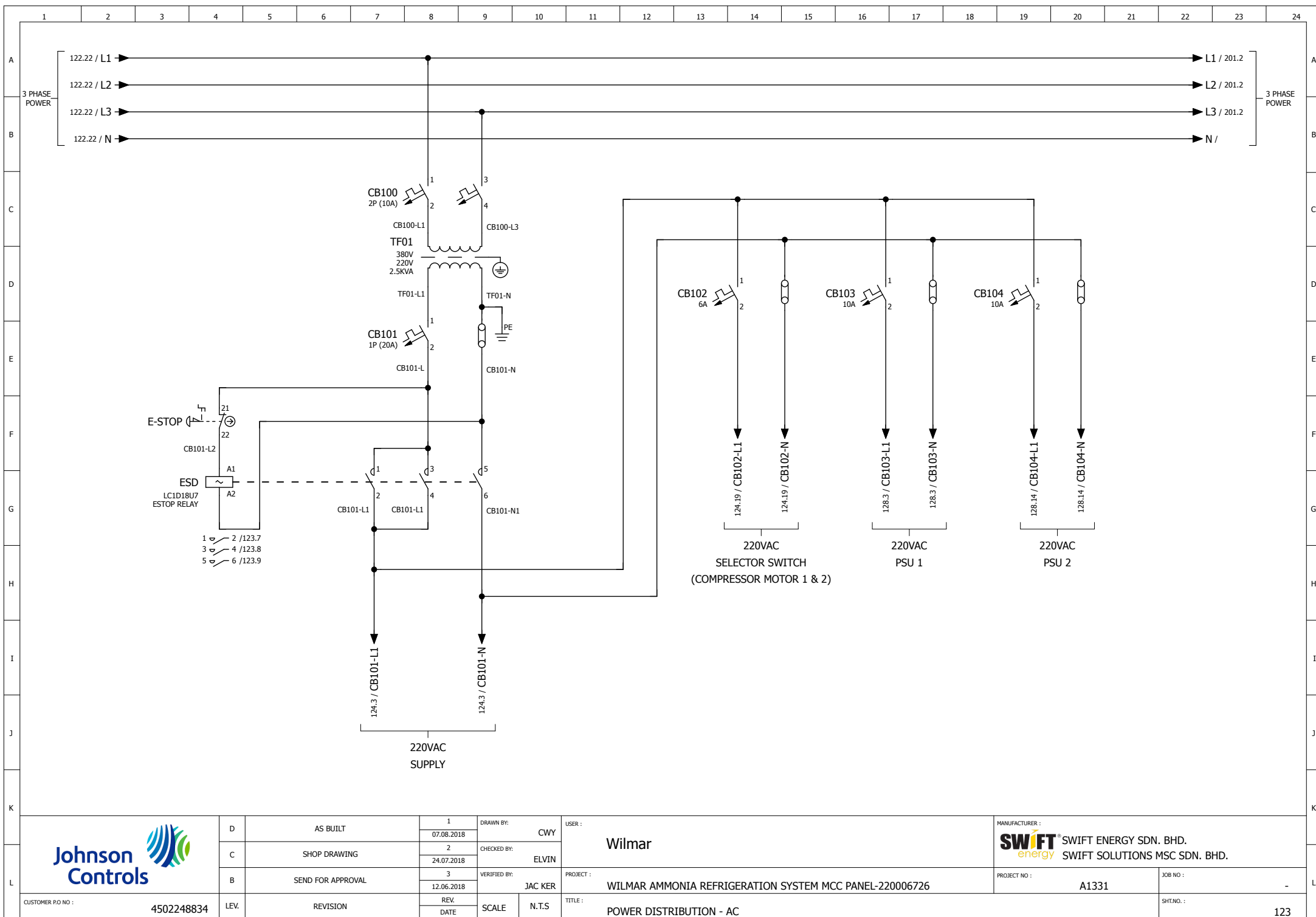
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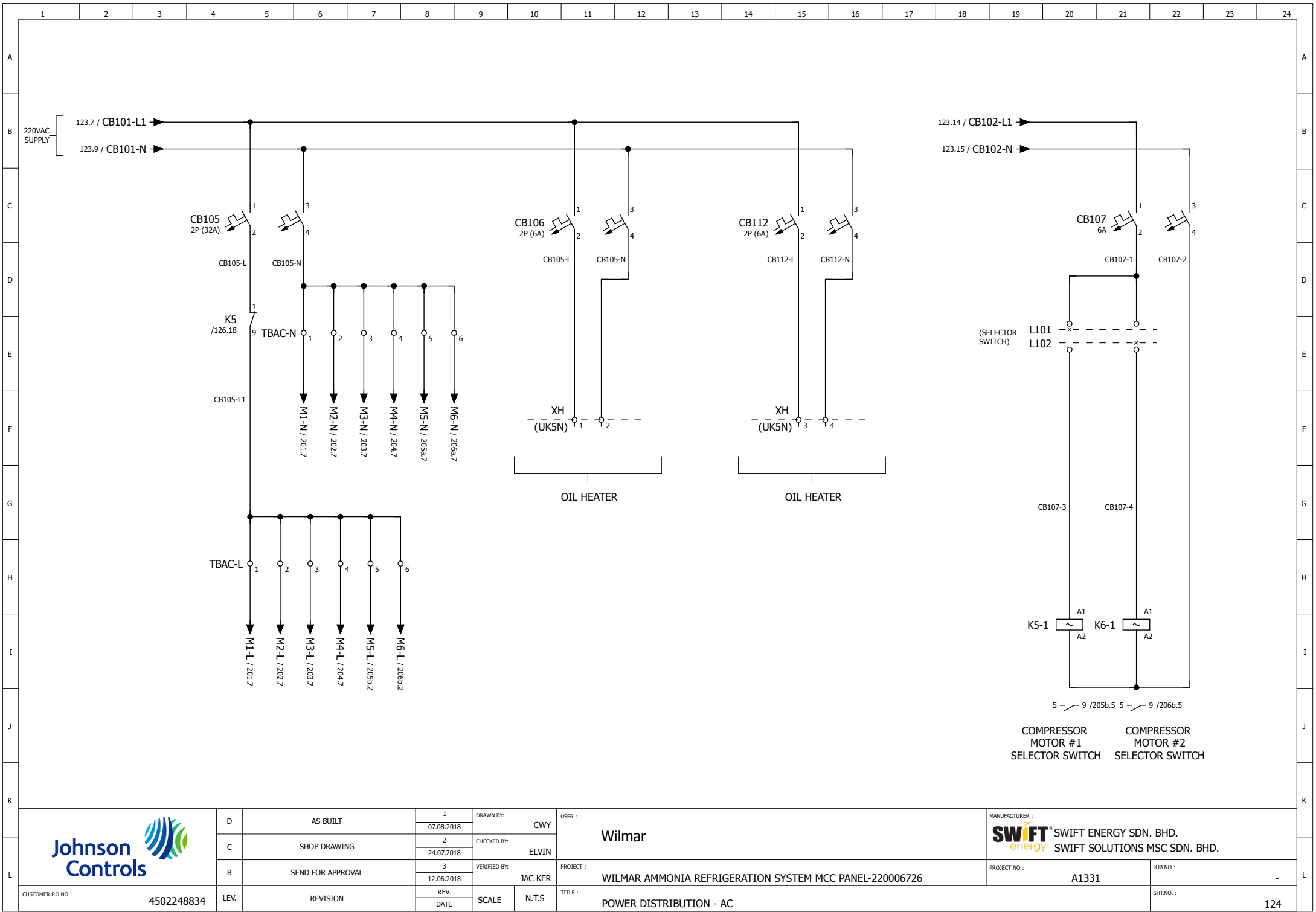
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PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	POWER DISTRIBUTION - AC

MANUFACTURER :	SWIFT energy	SWIFT ENERGY SDN. BHD.
		SWIFT SOLUTIONS MSC SDN. BHD.
PROJECT NO :	A1331	JOB NO :
		-
SHT.NO. :		121



CUSTOMER P.O NO : 4502248834	D	AS BUILT	1	DRAWN BY:	CWY	USER : Wilmar	MANUFACTURER : SWIFT ^{energy} SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
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	LEV.	REVISION	REV. DATE	SCALE	N.T.S			
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TITLE : POWER DISTRIBUTION - AC						SHT.NO. :		122



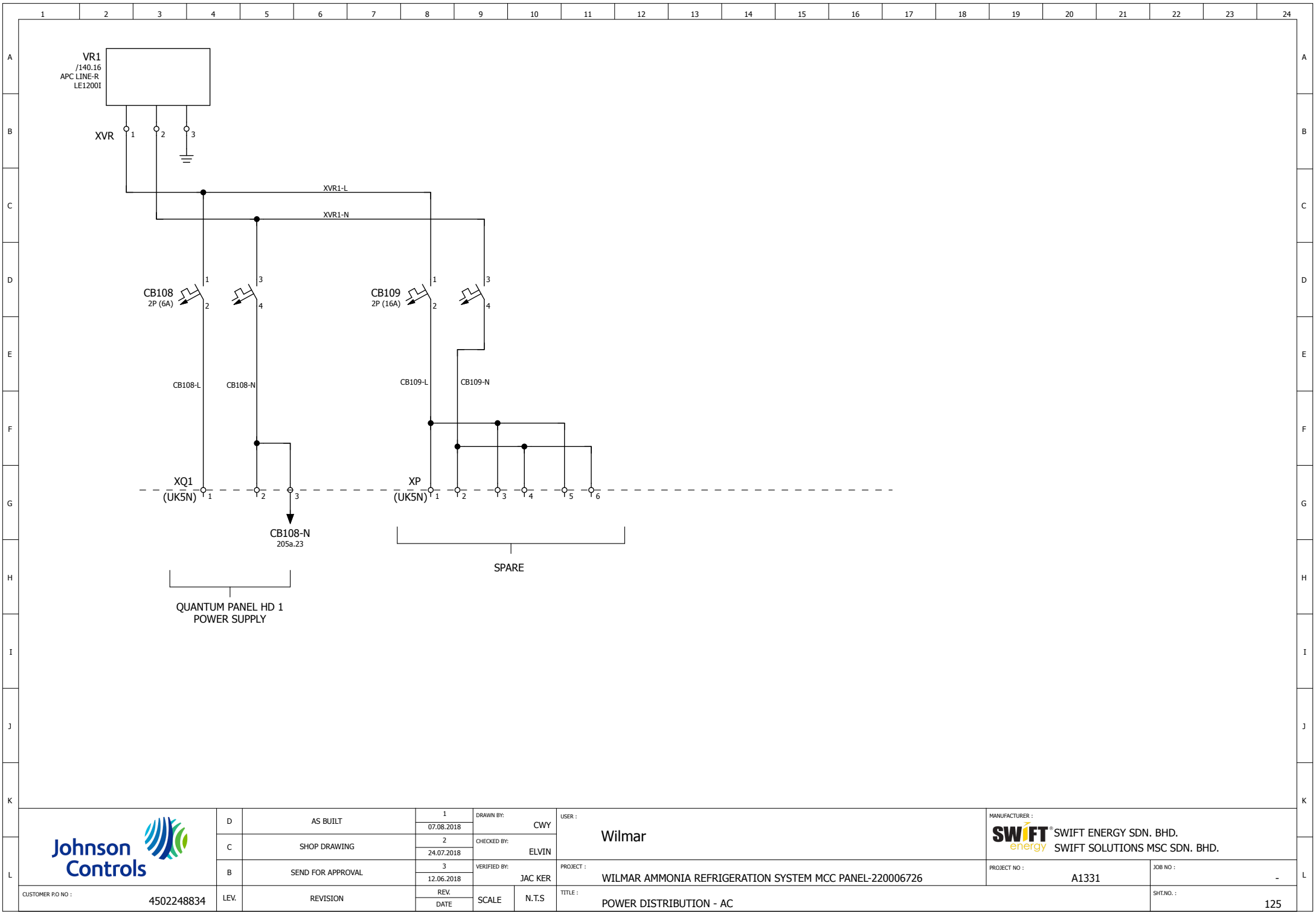


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LEV.	REVISION	REV.	SCALE	N.T.S
		DATE		

CUSTOMER P.O NO : 4502248834

USER :	Wilmar
PROJECT :	
WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726	
TITLE :	POWER DISTRIBUTION - AC

MANUFACTURER :		SWIFT energy SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
PROJECT NO :		A1331	JOB NO : -
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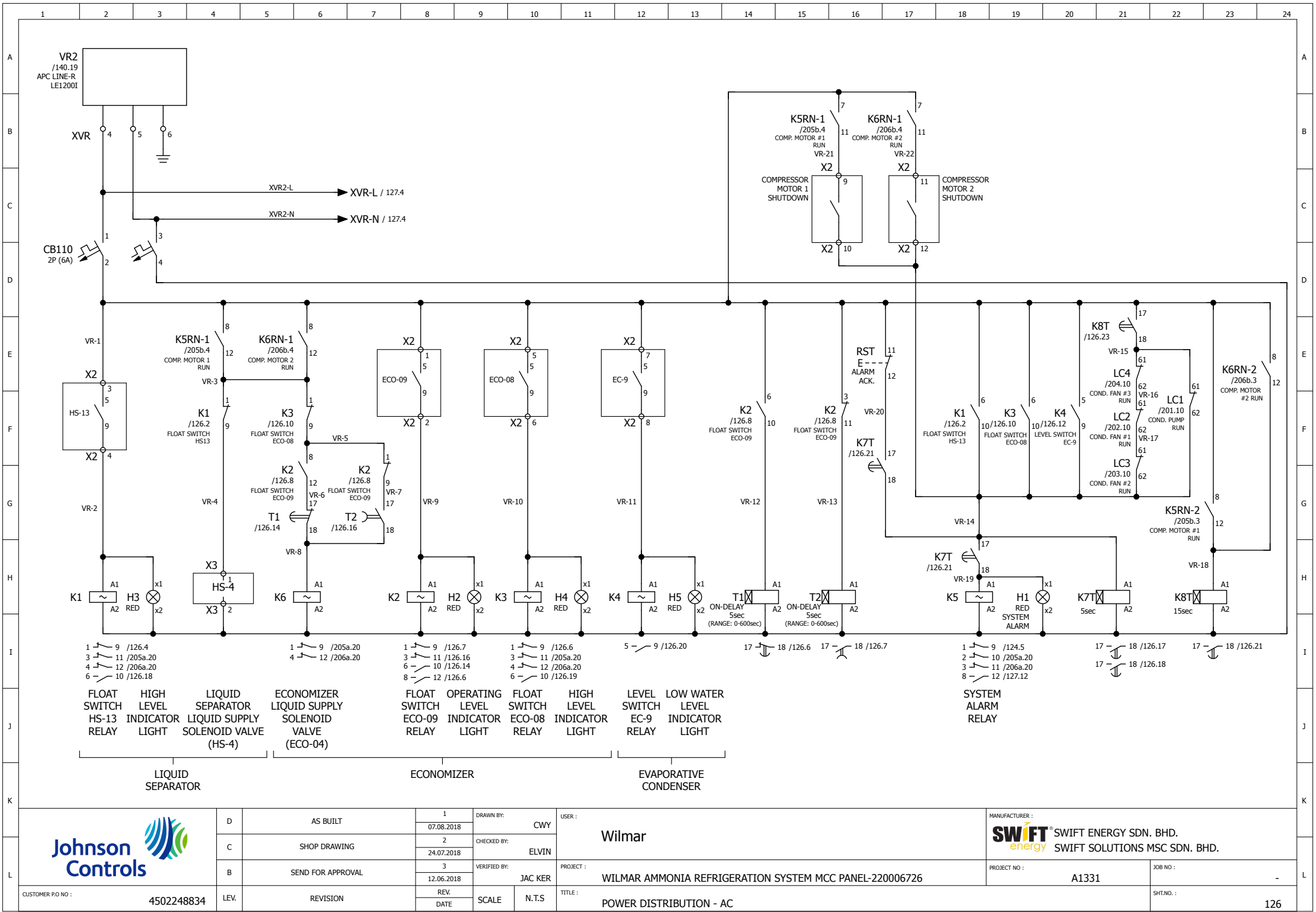


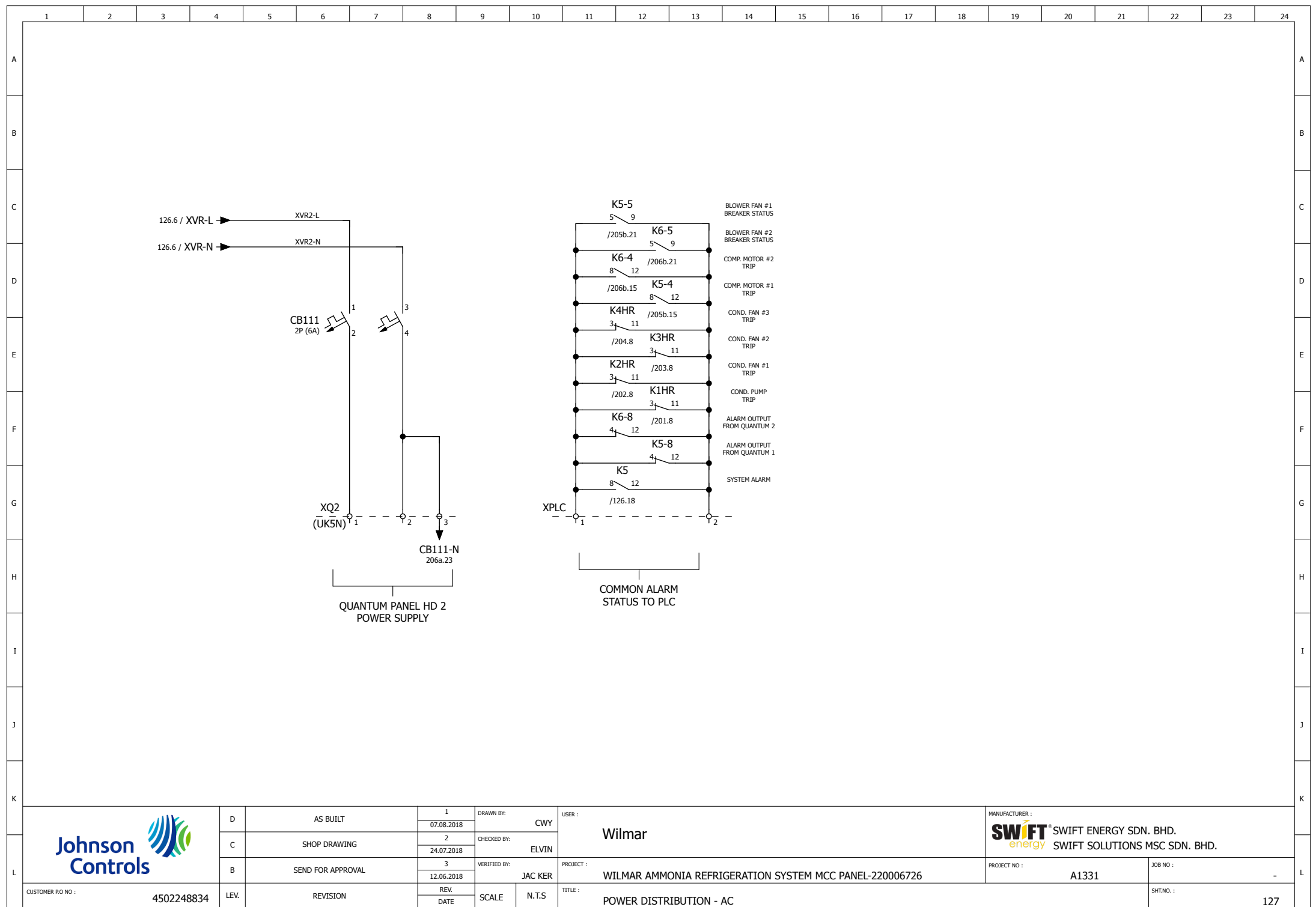
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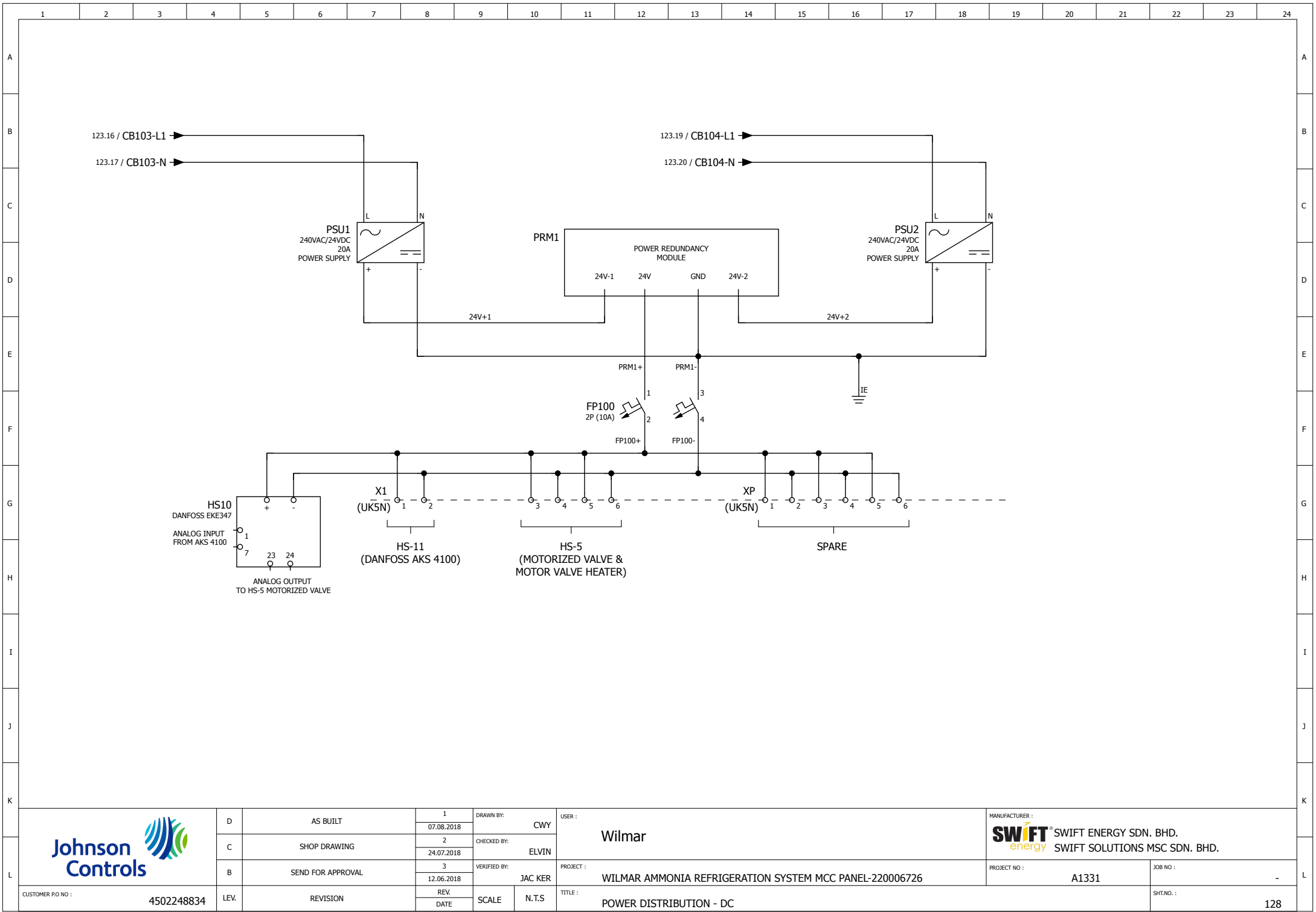
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PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	POWER DISTRIBUTION - AC

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SHT.NO. :	125

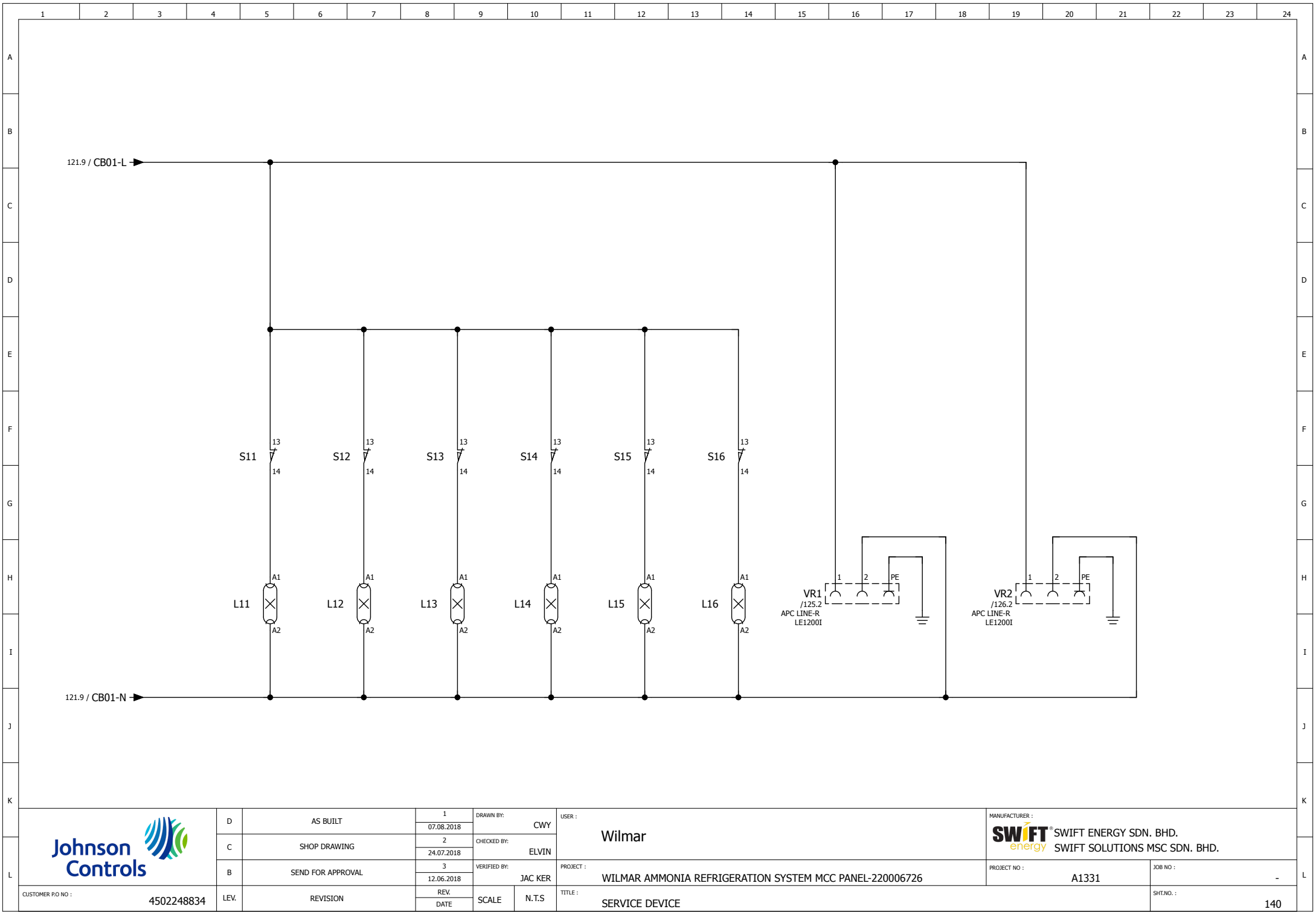
CUSTOMER P.O NO : 4502248834







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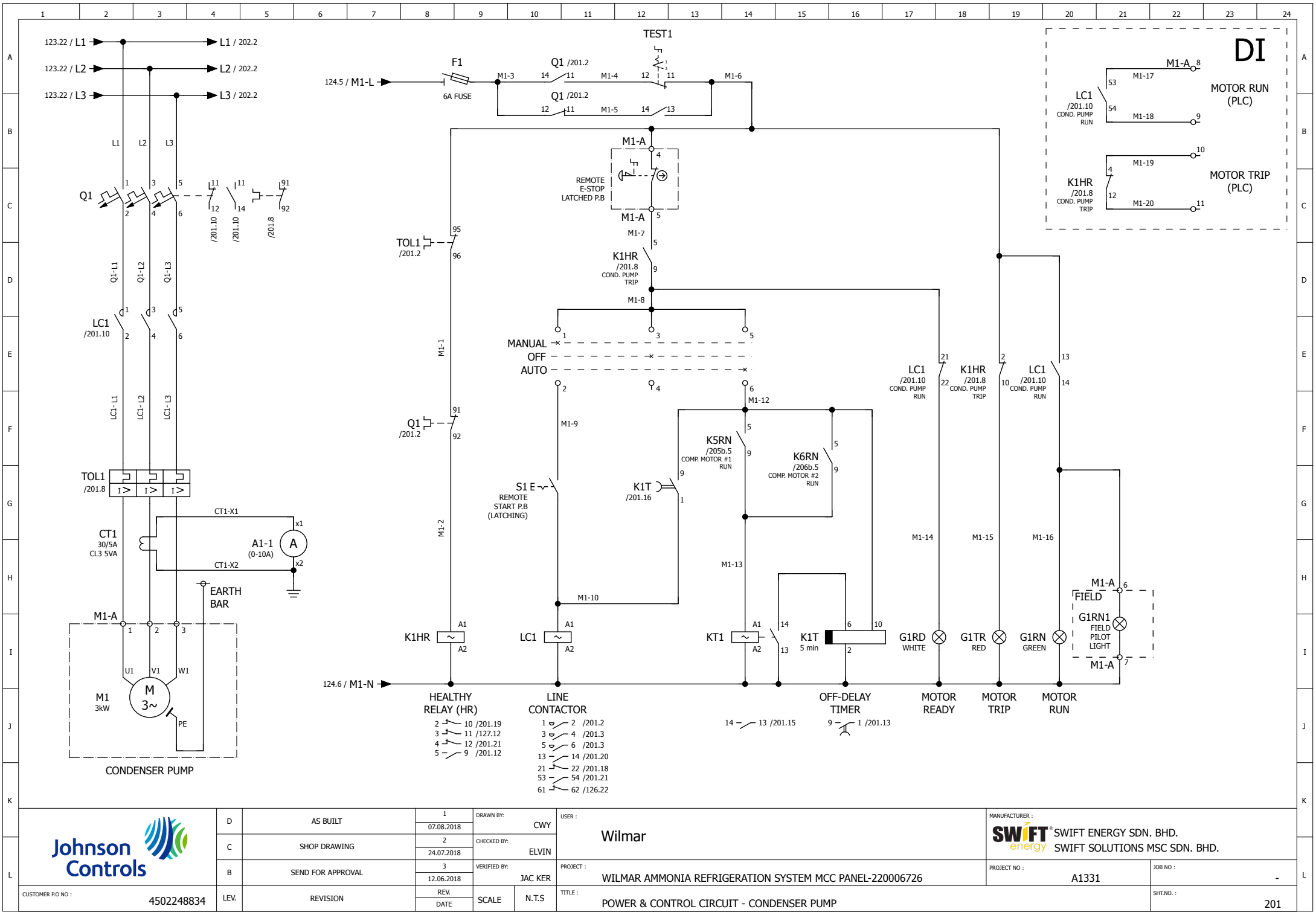
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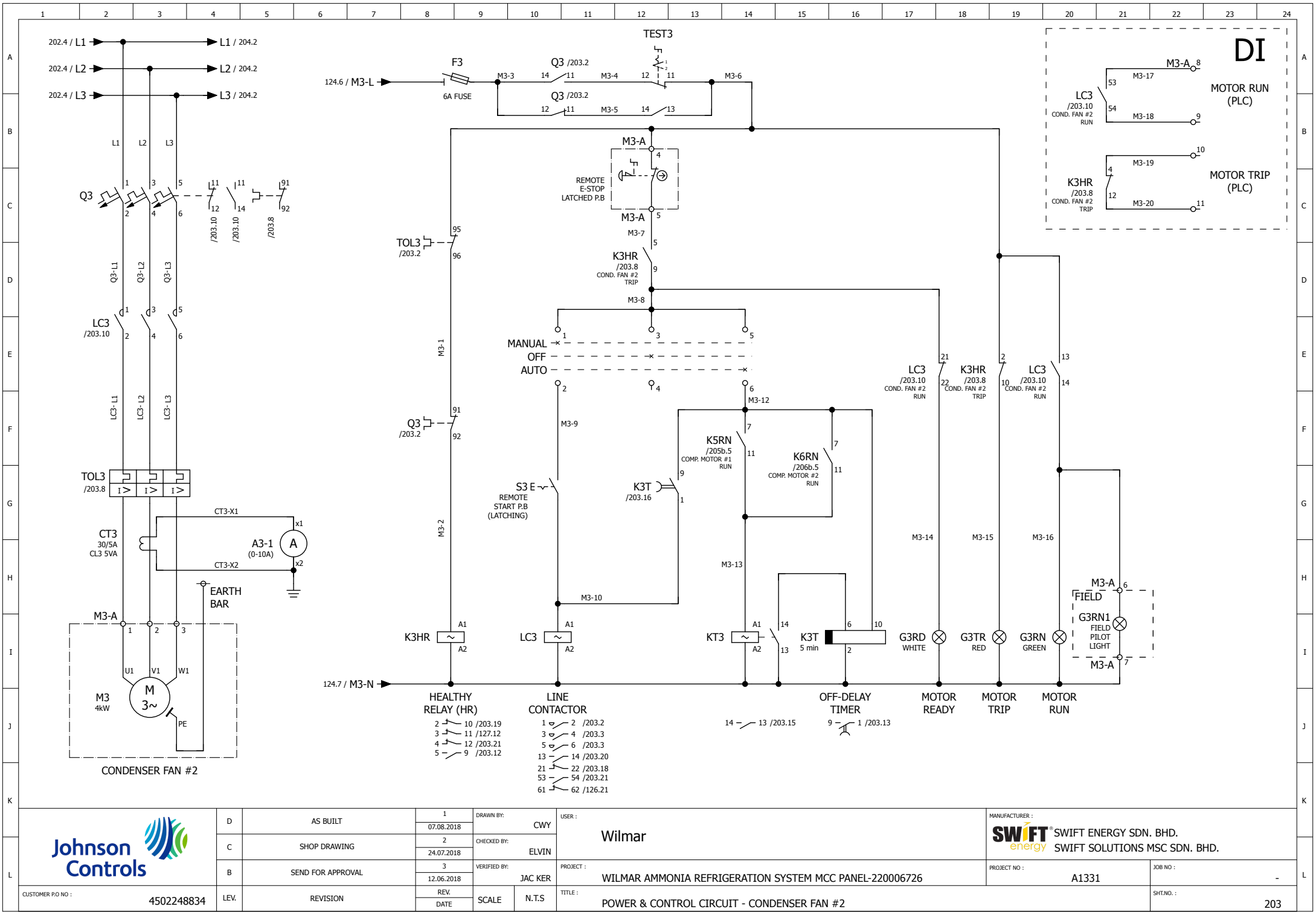
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PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	SERVICE DEVICE

MANUFACTURER :	
SWIFT [®] SWIFT ENERGY SDN. BHD. energy SWIFT SOLUTIONS MSC SDN. BHD.	
PROJECT NO :	JOB NO :
A1331	-
	SHT.NO. :
	140

CUSTOMER P.O NO : 4502248834

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	Type of Starter	Motor Size (A)	Motor Seq. No. (x)	Motor Tag (y)	Cable Size		Qx (Circuit Breaker)	LCx (Line Contactor)	DCx (Delta Contactor)	SCx (Star Contactor)	TOLx (Thermal Overload)	TBy (Terminal Block)												
					(B)	(C)																		
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D	TYPICAL VFD	315kW	5	COMPRESSOR MOTOR #1	FLEXIBLE BAR	FLEXIBLE BAR	3P MCCB 630A	-	-	-	-	-												
			6	COMPRESSOR MOTOR #2																				
E	TYPICAL DOL	2.2kW	5-2	BLOWER FAN #1	2.5mm²	2.5mm²	GV2-ME10 (4-6.3A)	-	-	-	-	UK5N												
			6-2	BLOWER FAN #2																				
F																								
G																								
H																								
I																								
J																								
K																								
	Johnson Controls 		D	AS BUILT	1	DRAWN BY: CWY		USER : Wilmar				MANUFACTURER : SWIFTenergy SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.												
C			SHOP DRAWING	2	CHECKED BY: ELVIN																			
B			SEND FOR APPROVAL	3	VERIFIED BY: JAC KER																			
					12.06.2018	PROJECT : WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726		PROJECT NO : A1331		JOB NO : -														
			CUSTOMER P.O NO : 4502248834	LEV.	REVISION	REV. DATE	SCALE	N.T.S	TITLE : TYPICAL STARTER TABLE					SHT.NO. : 141										

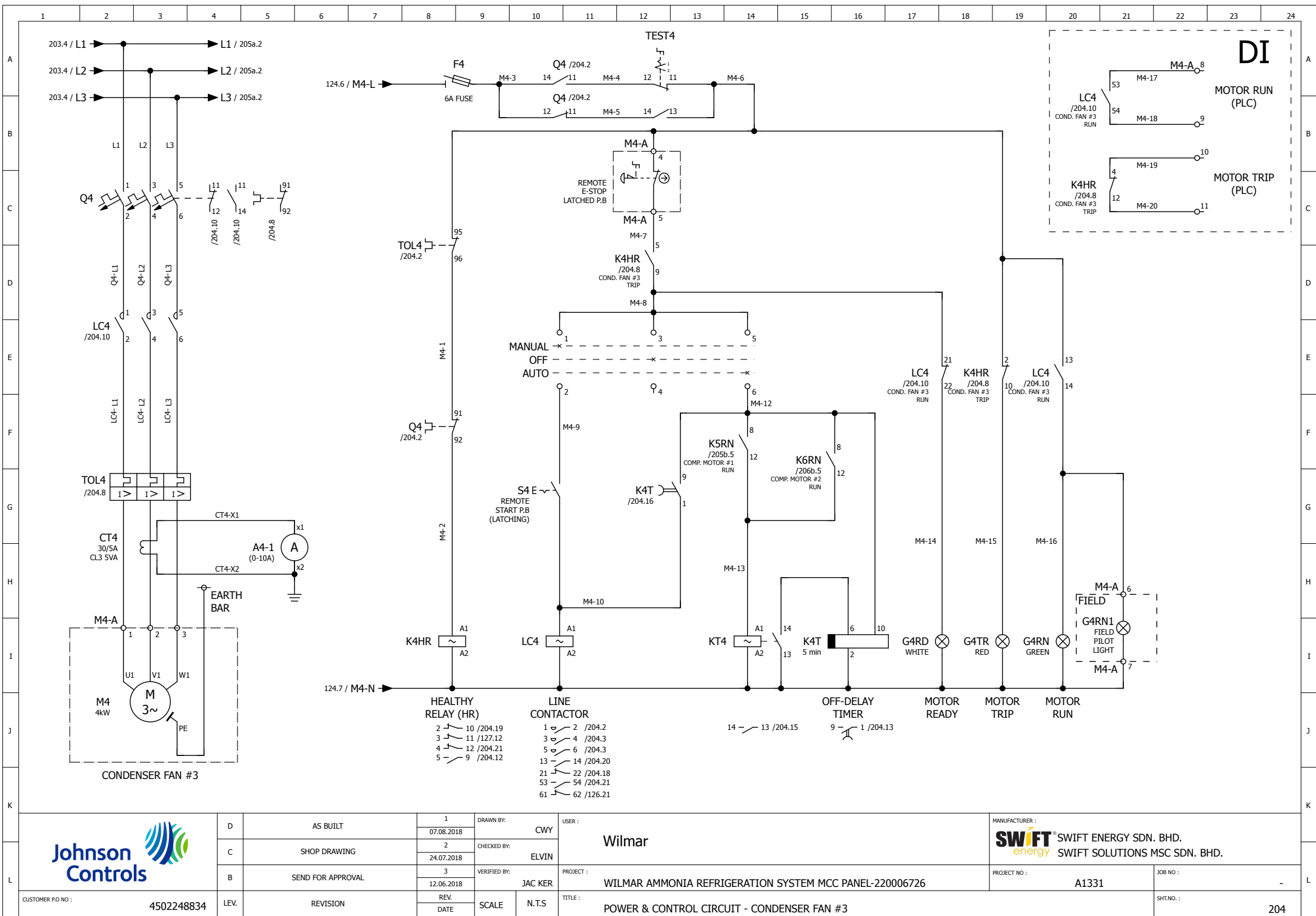




D	AS BUILT	1	DRAWN BY:	CWY
C	SHOP DRAWING	2	CHECKED BY:	ELVIN
B	SEND FOR APPROVAL	3	VERIFIED BY:	JAC KER
LEV.	REVISION	DATE	SCALE	N.T.S

USER :	WILMAR
PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	POWER & CONTROL CIRCUIT - CONDENSER FAN #2

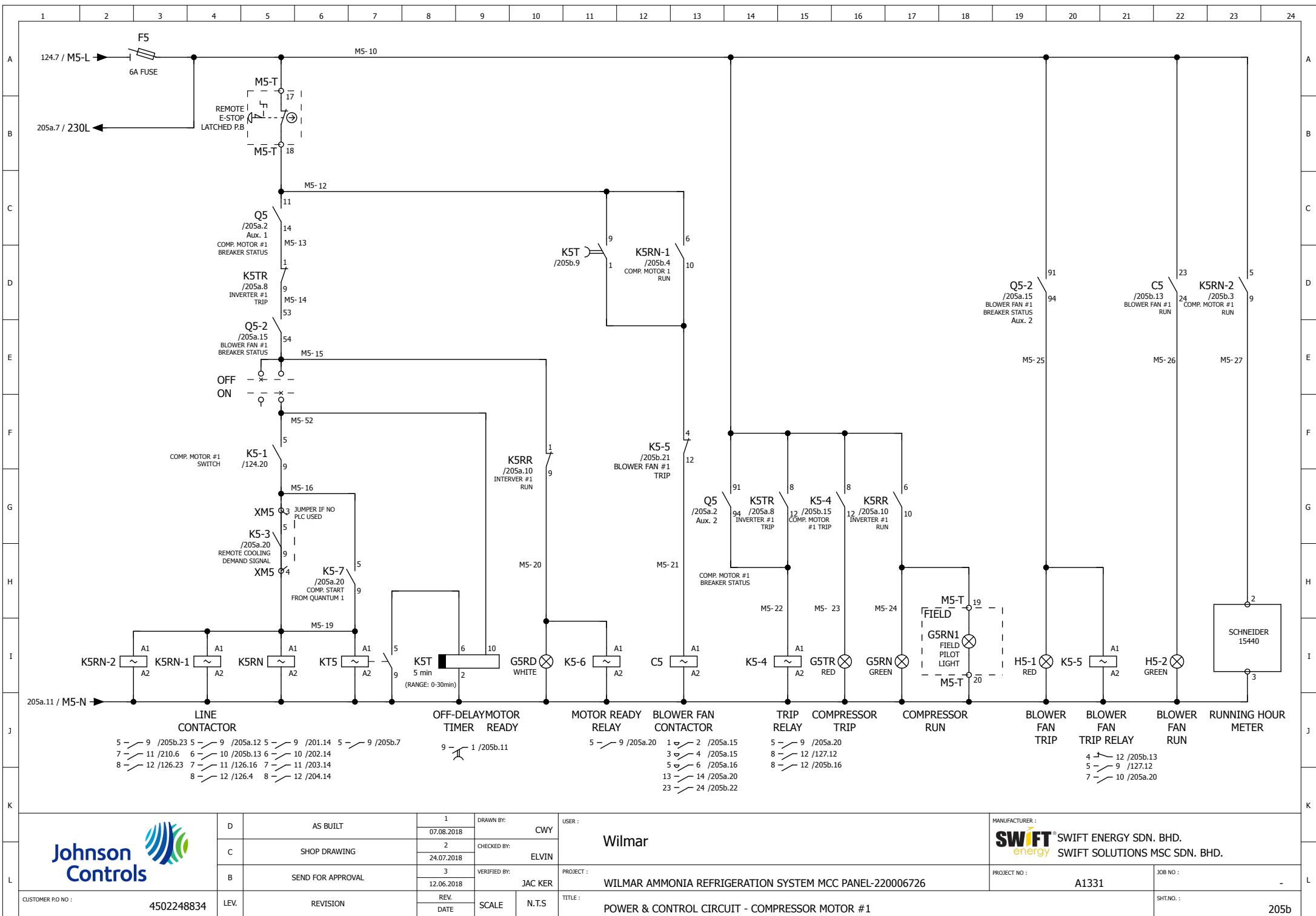
MANUFACTURER :	SWIFT energy	SWIFT ENERGY SDN. BHD.	PROJECT NO :	A1331	JOB NO :	-
		SWIFT SOLUTIONS MSC SDN. BHD.			SHT.NO. :	203

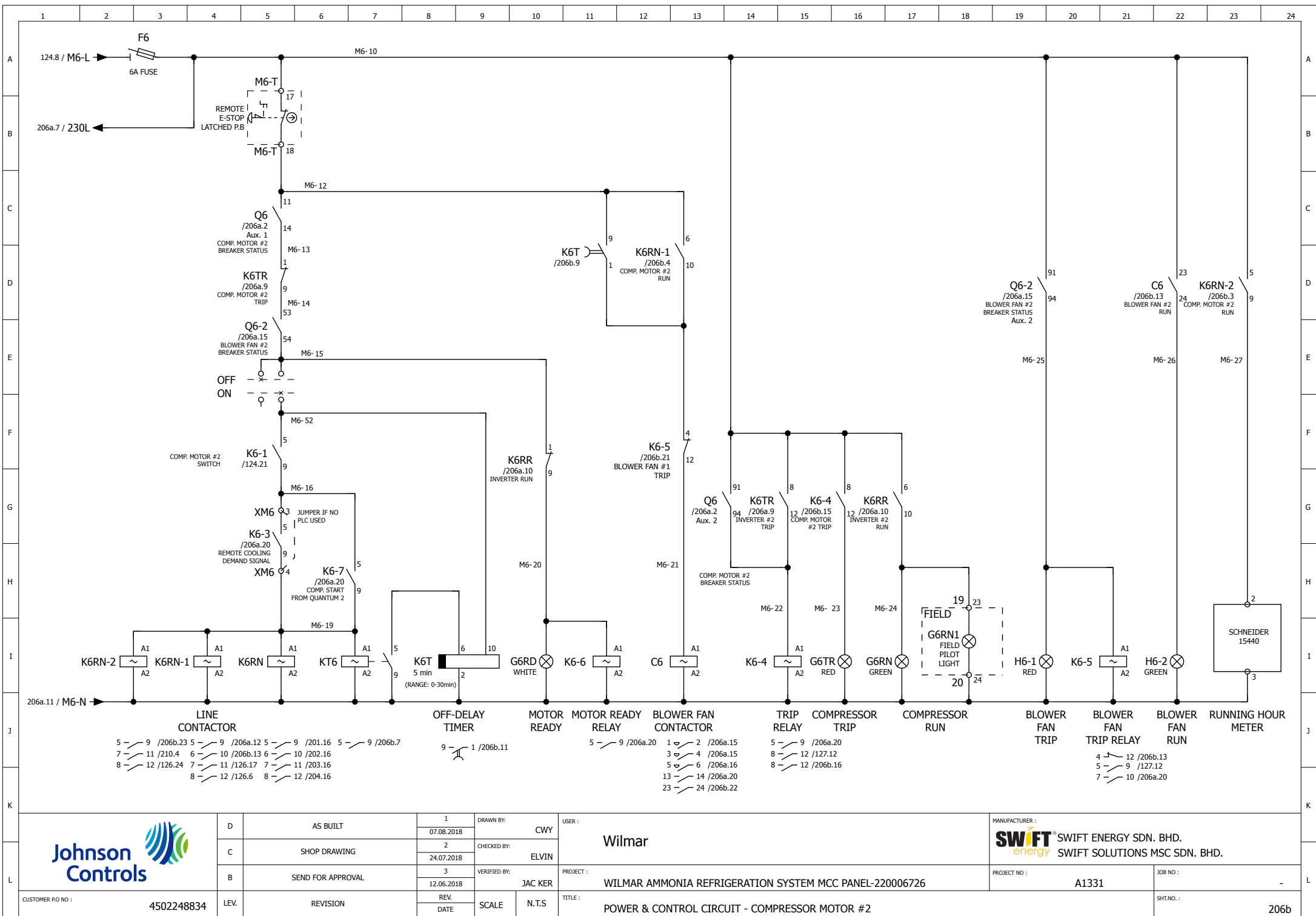


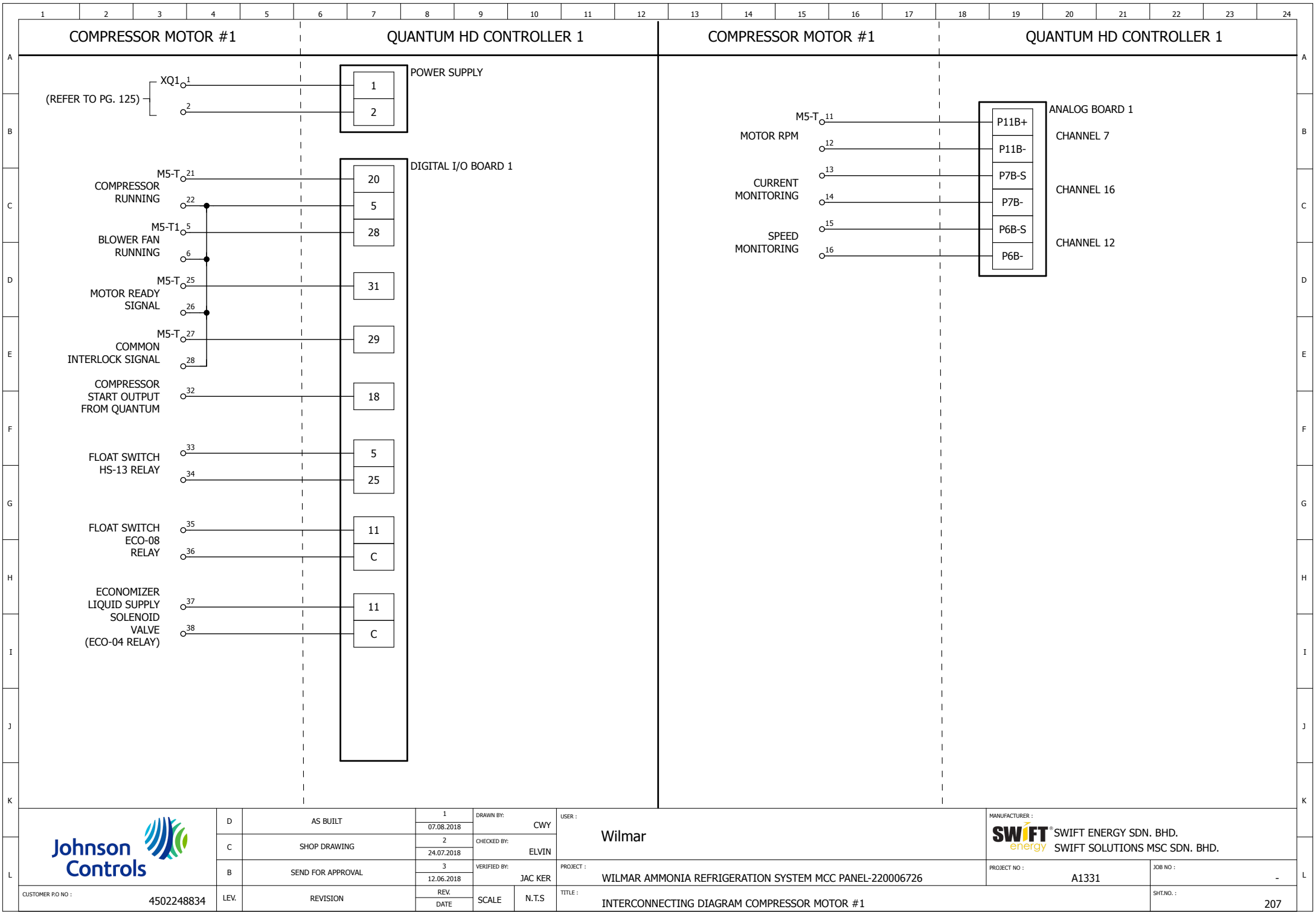
D	AS BUILT	1	DRAWN BY:	CWY
C	SHOP DRAWING	2	CHECKED BY:	ELVIN
B	SEND FOR APPROVAL	3	VERIFIED BY:	JAC KER
LEV.	REVISION	DATE	SCALE	N.T.S

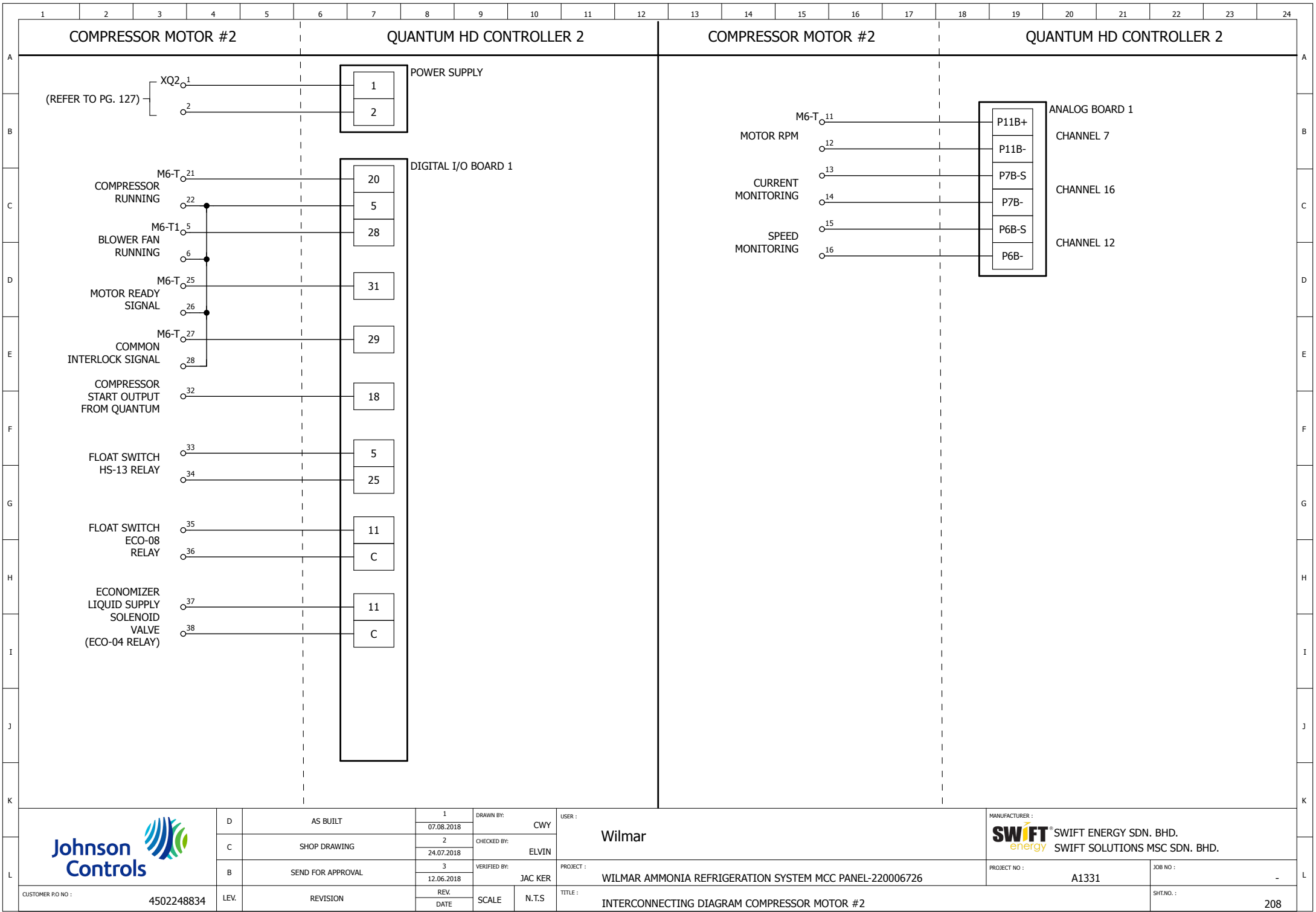
USER :	WILMAR
PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	POWER & CONTROL CIRCUIT - CONDENSER FAN #3

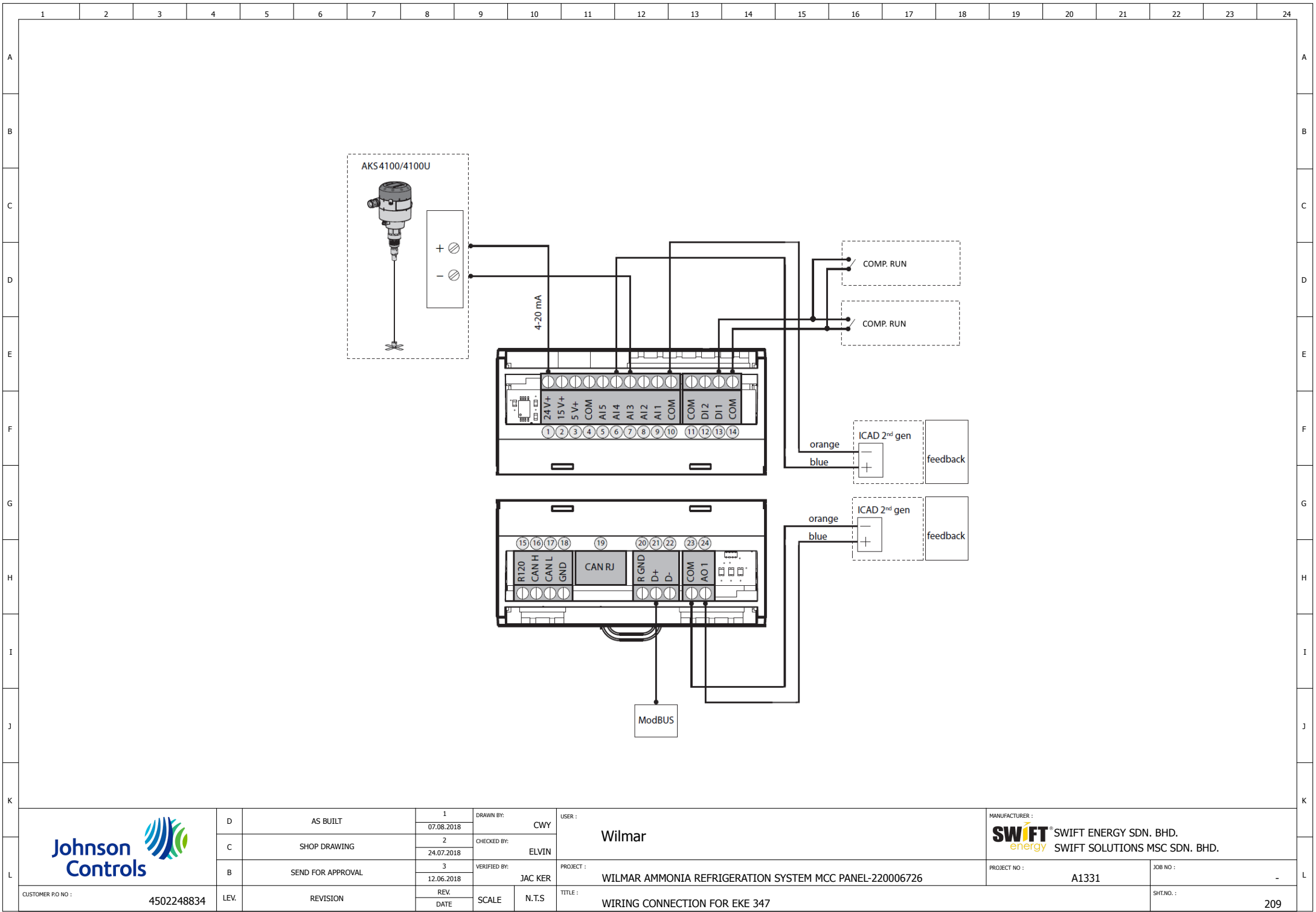
MANUFACTURER :	SWIFT energy	SWIFT ENERGY SDN. BHD.	PROJECT NO :	A1331	JOB NO :	-
		SWIFT SOLUTIONS MSC SDN. BHD.			SHT.NO. :	204











D	AS BUILT	1	DRAWN BY:	CWY
		07.08.2018		
C	SHOP DRAWING	2	CHECKED BY:	ELVIN
		24.07.2018		
B	SEND FOR APPROVAL	3	VERIFIED BY:	JAC KER
		12.06.2018		
LEV.	REVISION	REV.	SCALE	N.T.S
		DATE		

USER :	WILMAR
PROJECT :	WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726
TITLE :	WIRING CONNECTION FOR EKE 347

MANUFACTURER : SWIFT energy SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
PROJECT NO : A1331	JOB NO : -
SHT.NO. :	209

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PNU#	Parameter	Setting
0-02	Motor Speed Unit	[1]
1-20	Motor Power	315
1-22	Motor Voltage	380
1-23	Motor frequency	50
1-24	Motor Current	
1-29	Auto motor adjustment	0
1-90	Thermal Motor Protection	[2]
1-93	Thermistor Source	1 or 2
3-00	Reference Range	[0]
3-02	Minimum Ref	0
3-03	Maximum Ref	50
3-15	Reference Resource 1	[1]
3-41	Ramp up time 1	5
3-42	Ramp down time 1	5
4-12*	Output Frequency Low Limit	15 [0]
4-14*	Output Frequency high Limit	50
5-10	Term. No 18	[8]
5-40 [0]	Relay Output	[4]
5-40 [1]	Relay Output	[5]
6-12	Terminal 53 Low current	4
6-13	Terminal 53 High current	20

Johnson Controls 	D	AS BUILT	1	DRAWN BY:		USER : Wilmar	MANUFACTURER : SWIFTenergy SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.		
			07.08.2018	CWY					
	C	SHOP DRAWING	2	CHECKED BY:					
			24.07.2018	ELVIN		PROJECT : WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726	PROJECT NO : A1331	JOB NO : -	
	B	SEND FOR APPROVAL	3	VERIFIED BY:					
		12.06.2018	JAC KER						
CUSTOMER PO NO : 4502248834	LEV.	REVISION	REV. DATE	SCALE	N.T.S	TITLE : VSD SETPOINT		SHT.NO. :	

211

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24						
A	PART LISTS																							A						
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J																								J						
K																								K						
L	CUSTOMER P.O NO :		LEV.	REVISION	SCALE	N.T.S																		L						
							D		AS BUILT		1 07.08.2018	DRAWN BY: CWY	USER :						MANUFACTURER : SWIFT [®] SWIFT ENERGY SDN. BHD. energy SWIFT SOLUTIONS MSC SDN. BHD.											
							C		SHOP DRAWING		2 24.07.2018																			
							B		SEND FOR APPROVAL		3 12.06.2018	VERIFIED BY: JAC KER																		
											REV. DATE		PROJECT : WILMAR AMMONIA REFRIGERATION SYSTEM MCC PANEL-220006726						PROJECT NO : A1331			JOB NO : -								
													TITLE : PART LIST									SHTNO. : A1297/ 251								

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A	PART LISTS																							A
B																								
C																								
D																								
E																								
F																								
G																								
H																								
I																								
J																								
K																								
L																								

TAG	PART NUMBER	DESCRIPTION	MANUFACTURER	QUANTITY
CT5-1	330CT000335	MH MCT 30/5A CL3_5VA C/W BRACKET	MUN HEAN	1
CT6	330CT000259	MH MCT 600/5A CL3_5VA C/W BRACKET	MUN HEAN	1
CT6-1	330CT000335	MH MCT 30/5A CL3_5VA C/W BRACKET	MUN HEAN	1
ESD	LC1D08U7	3P CONTACTOR COIL VOLTAGE 240V + Auxiliary Contact 1 N/O + 1 N/C	SCHNEIDER ELECTRIC	1
E-STOP	ZBZ1604	EMERGENCY STOP PB MUSHROOM GUARD RED	SCHNEIDER ELECTRIC	1
E-STOP	ZB5AS844	22mm RED Emergency pushbutton head latching turn release	SCHNEIDER ELECTRIC	1
E-STOP	ZB5AZ102	SINGLE NO FLUSH PUSH BUTTON BODY - BLACK	SCHNEIDER ELECTRIC	1
F01	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	3
F02	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	3
F03	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F1	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F2	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F3	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F4	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F5	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
F6	A9N15636	SCH 1 to 32A 1P 10.3x38 ISOLATOR FUSE-CARRIER W/O FUSE	SCHNEIDER ELECTRIC	1
FP100	A9F74210	2P 10 10KA	SCHNEIDER ELECTRIC	1
G1RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1
G1RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1
G1TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1
G2RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1
G2RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1
G2TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1
G3RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1
G3RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1
G3TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1
G4RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1
G4RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1
G4TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1
G5RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1

	D	AS BUILT	1 07.08.2018	DRAWN BY: CWY	USER :	 SWIFT ENERGY SDN. BHD. SWIFT SOLUTIONS MSC SDN. BHD.	
	C	SHOP DRAWING	2 24.07.2018	CHECKED BY: ELVIN			
	B	SEND FOR APPROVAL	3 12.06.2				

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24																																																																																																																																																											
PART LISTS																																																																																																																																																																																		
<table><tr><th>TAG</th><th>PART NUMBER</th><th>DESCRIPTION</th><th>MANUFACTURER</th><th>QUANTITY</th></tr><tr><td>G5RN</td><td>SMT22240VACG</td><td>SHIMATEC 240V AC PILOT LIGHT GREEN</td><td>SHIMATEC</td><td>1</td></tr><tr><td>G5TR</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>G6RD</td><td>390SSW000047</td><td>SHIMATEC 240V AC PILOT LIGHT WHITE</td><td>SHIMATEC</td><td>1</td></tr><tr><td>G6RN</td><td>SMT22240VACG</td><td>SHIMATEC 240V AC PILOT LIGHT GREEN</td><td>SHIMATEC</td><td>1</td></tr><tr><td>G6TR</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H1</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H2</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H3</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H4</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H5</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H5-1</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H5-2</td><td>SMT22240VACG</td><td>SHIMATEC 240V AC PILOT LIGHT GREEN</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H6-1</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr><tr><td>H6-2</td><td>SMT22240VACG</td><td>SHIMATEC 240V AC PILOT LIGHT GREEN</td><td>SHIMATEC</td><td>1</td></tr><tr><td>K1</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K1HR</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K2</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K2HR</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K3</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K3HR</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K4</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K4HR</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-1</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-3</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-4</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-5</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-6</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5-7</td><td>RXM2AB2P7</td><td>SCH 2P 230V AC AC MINIATURE RELAYS WITH LED</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr><tr><td>K5RN</td><td>RXM4AB2P7</td><td>4P RELAY 230VAC</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr></table>																								TAG	PART NUMBER	DESCRIPTION	MANUFACTURER	QUANTITY	G5RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1	G5TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	G6RD	390SSW000047	SHIMATEC 240V AC PILOT LIGHT WHITE	SHIMATEC	1	G6RN	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1	G6TR	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H1	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H2	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H3	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H4	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H5	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H5-1	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H5-2	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1	H6-1	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	H6-2	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1	K1	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K1HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K2	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K2HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K3	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K3HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K4	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K4HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K5	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K5-1	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K5-3	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K5-4	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K5-5	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1	K5-6	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K5-7	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1	K5RN	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1
TAG	PART NUMBER	DESCRIPTION	MANUFACTURER	QUANTITY																																																																																																																																																																														
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H2	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H3	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H4	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H5	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H5-1	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H5-2	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1																																																																																																																																																																														
H6-1	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1																																																																																																																																																																														
H6-2	SMT22240VACG	SHIMATEC 240V AC PILOT LIGHT GREEN	SHIMATEC	1																																																																																																																																																																														
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K1HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
K2	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
K2HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
K3	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
K3HR	RXM2AB2P7	SCH 2P 230V AC AC MINIATURE RELAYS WITH LED	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
K4	RXM4AB2P7	4P RELAY 230VAC	SCHNEIDER ELECTRIC	1																																																																																																																																																																														
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L	<table border="1"> <thead> <tr> <th>TAG</th><th>PART NUMBER</th><th>DESCRIPTION</th><th>MANUFACTURER</th><th>QUANTITY</th></tr> </thead> <tbody> <tr><td>PM8240</td><td>METSEPM8240</td><td>PowerLogic Panel mount meter</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>PRM1</td><td>6EP1961-3BA21</td><td>SITOP PSE202U REDUNDANCY MODULE</td><td>SIEMENS</td><td>1</td></tr> <tr><td>PSU1</td><td>6EP1336-2BA10</td><td>SITOP PSU100S 20 A stabilized power supply</td><td>SIEMENS</td><td>1</td></tr> <tr><td>PSU2</td><td>6EP1336-2BA10</td><td>SITOP PSU100S 20 A stabilized power supply</td><td>SIEMENS</td><td>1</td></tr> <tr><td>Q1</td><td>A9F74316</td><td>3P MCB 16A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q2</td><td>A9F74320</td><td>3P MCB 20A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q3</td><td>A9F74320</td><td>3P MCB 20A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q4</td><td>A9F74320</td><td>3P MCB 20A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q5</td><td>LV432876</td><td>3P MCCB 630A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q5-2</td><td>GV2ME10</td><td>SCH GV2ME 4-6.3 THERMAL MAGNETIC MCB</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q6</td><td>LV432876</td><td>3P MCCB 630A</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>Q6-2</td><td>GV2ME10</td><td>SCH GV2ME 4-6.3 THERMAL MAGNETIC MCB</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>R</td><td>SMT22240VACR</td><td>SHIMATEC 240V AC PILOT LIGHT RED</td><td>SHIMATEC</td><td>1</td></tr> <tr><td>RHM1</td><td>15440</td><td>RUNNING HOUR METER</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>RHM2</td><td>15440</td><td>RUNNING HOUR METER</td><td>SCHNEIDER ELECTRIC</td><td>1</td></tr> <tr><td>RST</td><td>APB22BL</td><td>BLACK PUSH BUTTON</td><td>CIKACHI</td><td>1</td></tr> <tr><td>S1</td><td>YW1B-A1E10G</td><td>REMOTE START</td><td>IDEC</td><td>1</td></tr> <tr><td>S11</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S12</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S13</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S14</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S15</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S16</td><td>HY-LS803N</td><td>HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH</td><td>HANYOUNG</td><td>1</td></tr> <tr><td>S2</td><td>YW1B-A1E10G</td><td>REMOTE START</td><td>IDEC</td><td>1</td></tr> <tr><td>S3</td><td>YW1B-A1E10G</td><td>REMOTE START</td><td>IDEC</td><td>1</td></tr> <tr><td>S4</td><td>YW1B-A1E10G</td><td>REMOTE START</td><td>IDEC</td><td>1</td></tr> <tr><td>S5</td><td>APB22BL</td><td>CKC APB-22 PUSH BUTTON - BLACK</td><td>CIKACHI</td><td>1</td></tr> <tr><td>S6</td><td>APB22BL</td><td>CKC APB-22 PUSH BUTTON - BLACK</td><td>CIKACHI</td><td>1</td></tr> <tr><td>SS1</td><td>S1661151-A</td><td>SALZER 16A 1 POLE A/O</td></tr></tbody></table>	TAG	PART NUMBER	DESCRIPTION	MANUFACTURER	QUANTITY	PM8240	METSEPM8240	PowerLogic Panel mount meter	SCHNEIDER ELECTRIC	1	PRM1	6EP1961-3BA21	SITOP PSE202U REDUNDANCY MODULE	SIEMENS	1	PSU1	6EP1336-2BA10	SITOP PSU100S 20 A stabilized power supply	SIEMENS	1	PSU2	6EP1336-2BA10	SITOP PSU100S 20 A stabilized power supply	SIEMENS	1	Q1	A9F74316	3P MCB 16A	SCHNEIDER ELECTRIC	1	Q2	A9F74320	3P MCB 20A	SCHNEIDER ELECTRIC	1	Q3	A9F74320	3P MCB 20A	SCHNEIDER ELECTRIC	1	Q4	A9F74320	3P MCB 20A	SCHNEIDER ELECTRIC	1	Q5	LV432876	3P MCCB 630A	SCHNEIDER ELECTRIC	1	Q5-2	GV2ME10	SCH GV2ME 4-6.3 THERMAL MAGNETIC MCB	SCHNEIDER ELECTRIC	1	Q6	LV432876	3P MCCB 630A	SCHNEIDER ELECTRIC	1	Q6-2	GV2ME10	SCH GV2ME 4-6.3 THERMAL MAGNETIC MCB	SCHNEIDER ELECTRIC	1	R	SMT22240VACR	SHIMATEC 240V AC PILOT LIGHT RED	SHIMATEC	1	RHM1	15440	RUNNING HOUR METER	SCHNEIDER ELECTRIC	1	RHM2	15440	RUNNING HOUR METER	SCHNEIDER ELECTRIC	1	RST	APB22BL	BLACK PUSH BUTTON	CIKACHI	1	S1	YW1B-A1E10G	REMOTE START	IDEC	1	S11	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S12	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S13	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S14	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S15	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S16	HY-LS803N	HANYOUNG HY-LS803N PUSH PLUNGER LIMIT SWITCH	HANYOUNG	1	S2	YW1B-A1E10G	REMOTE START	IDEC	1	S3	YW1B-A1E10G	REMOTE START	IDEC	1	S4	YW1B-A1E10G	REMOTE START	IDEC	1	S5	APB22BL	CKC APB-22 PUSH BUTTON - BLACK	CIKACHI	1	S6	APB22BL	CKC APB-22 PUSH BUTTON - BLACK	CIKACHI	1	SS1	S1661151-A	SALZER 16A 1 POLE A/O
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